Code No: 136BC **R16**

JAWAHARLAL NEHRU TECHNOLOGICAL UNIVERSITY HYDERABAD
B. Tech III Year II Semester Examinations, July - 2023
DESIGN PATTERNS
(Computer Science and Engineering)

Time: 3 Hours **Max. Marks: 75**

Note: i) Question paper consists of Part A, Part B.
ii) Part A is compulsory, which carries 25 marks. In Part A, Answer all questions.
iii) In Part B, Answer any one question from each unit. Each question carries 10 marks and may have a, b as sub questions.

Used paper July/Aug-2023

PART – A

(25 Marks)

1.a)	Explain how to select a design pattern.	[2]
b)	Write a brief note on the catalog of design patterns.	[3]
c)	What is Formatting?	[2]
d)	Write a short note on Document Structure.	[3]
e)	Write a short note on Abstract Factory.	[2]
f)	Write a brief note on Singleton.	[3]
g)	Write a short note on Façade.	[2]
h)	Briefly explain about Bridge.	[3]
i)	Discuss briefly about State.	[2]
j)	Write a short note on Chain of Responsibility.	[3]

PART – B

(50 Marks)

2.	What is a Design Pattern? What are elements of design pattern? Explain in detail about design patterns in Smalltalk MVC.	[10]
	OR	
3.a)	Discuss about describing Design Patterns.	
b)	Write a brief note on how to use a design pattern.	[5+5]
4.	Write a detailed note on supporting Multiple Look-and-Feel standards.	[10]
	OR	
5.	Discuss in detail about Supporting Multiple Window Systems.	[10]
6.	Write a detailed note on Factory Method.	[10]
	OR	
7.a)	Write a short note on Prototype.	
b)	Elaborate discussion of Creational Patterns.	[5+5]
8.a)	Discuss in detail about Decorator.	
b)	Write a short note on Adapter.	[5+5]
	OR	
9.a)	Explain in detail about Proxy.	
b)	Explain briefly about Composite.	[5+5]

10.a) Discuss in detail about Mediator.

b) Write a detailed note on Visitor. [5+5]

OR

11. Explain in detail about the following Behavioral Patterns:

a) Iterator

b) Strategy. [5+5]

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